

# Fajar Adie Santosa

✉ fajaradios01@gmail.com | 🌐 fajardios.my.id | 🔗 LinkedIn | 🐙 GitHub

## SUMMARY

Data Scientist with more than one year of experience in data analysis, machine learning, and AI integration. Experienced in collecting, cleaning, validating, and analyzing multi-source data, including **7,601 school records**, to produce reproducible analyses and decision-ready insights. Able to integrate data and AI workflows into backend systems using Python, Laravel, FastAPI, REST APIs, SQL databases, OCR, and LLM APIs.

## EXPERIENCE

### PT Sentra Vidya Utama (SEVIMA)

Feb. 2025 - Aug. 2025

*Data Analyst Intern*

Surabaya, Indonesia

- Collected and structured **7,601 school records** from Dapodik and public school/foundation websites to support lead generation and market expansion.
- Filtered 2,835 private-school records into **523 eligible prospects**, then cleaned and manually labeled tuition data for 105 schools while documenting assumptions and data limitations.
- Helped the administration team design an OCR-assisted workflow for collecting and consolidating data from PDF and Excel files, reducing processing time by approximately **50%** while producing cleaner, consistently structured data from multiple teams.
- Translated findings into a customer-targeting analysis that identified **5 prospective customers**, produced weekly reports, and documented each analysis and its findings for marketing teams.

### Institut Teknologi Sepuluh Nopember

Aug. 2024 - Sept. 2025

*Programming Laboratory Teaching Assistant*

Surabaya, Indonesia

- Mentored more than **20 Mathematics students** in Java, programming fundamentals, and object-oriented problem solving.
- Prepared weekly laboratory exercises, assessments, and clear learning documentation, then reviewed student work and explained corrections.
- Developed JUnit test suites to automate code reviews and verify the correctness of students' solutions, enabling faster and more consistent assessment.

## SELECTED PROJECTS

### Capturo Document OCR Platform

2026

*Laravel, FastAPI, PaddleOCR, PostgreSQL*

- Built an end-to-end platform that converts invoices, forms, and spreadsheets into structured records using template-based extraction, batch upload, bounding-box and confidence previews, manual review, and Excel export.
- Designed **12 Laravel REST endpoints** and user-scoped data, OCR mapping, multi-page geometry, and batch isolation.

### Physics-Informed Neural Network Sparsity Research

2026

*Python, TensorFlow, SciPy, NumPy, Matplotlib*

- Developed Physics-Informed Neural Networks to solve the one-dimensional Poisson and Bratu equations and a damped harmonic oscillator by incorporating governing equations and boundary conditions into the training objective.
- Compared dense and weight-sparse PINN architectures across solution accuracy, parameter count, and computational efficiency.

### FinSentry Financial Anomaly Monitoring Agent

2026

*FastAPI, PostgreSQL, Next.js, Hermes Agent, MCP, Docker*

- Architected a local-first monitoring MVP that combines a deterministic risk engine, Telegram verification workflow, and Hermes Agent with exactly **four permission-scoped MCP tools**.
- Implemented idempotent event ingestion, a transactional outbox, offline retry, signed sessions, local TLS, and encrypted PostgreSQL backup/restore.

## EDUCATION

**Institut Teknologi Sepuluh Nopember (ITS)**  
*Bachelor of Mathematics (Computer Science Field)*

Aug. 2022 - Aug. 2026  
Surabaya, Indonesia

- GPA: **3.19/4.00** | EFL-ITS: **477**
- **Relevant Courses:** Algorithms and Programming, Statistical Methods, Numerical Methods, Data Science, Linear Algebra, and Introduction to Numerical Optimization.
- **Thesis title :** *The Effect of Weight Sparsity on The Accuracy and Efficiency of Physics-Informed Neural Networks*  
**Outcome :** Reduced the number of model parameters through weight sparsity while maintaining accuracy comparable to dense neural networks.

## SKILLS

|                                     |                                                                         |
|-------------------------------------|-------------------------------------------------------------------------|
| <b>Programming Languages</b>        | Python, SQL, Java, JavaScript, C++, and MATLAB                          |
| <b>Backend &amp; AI Integration</b> | Laravel, FastAPI, REST APIs, JSON, OpenAI API, LLM integration, and OCR |
| <b>Databases</b>                    | PostgreSQL and MySQL                                                    |
| <b>Tools &amp; Deployment</b>       | Docker, Git/GitHub, Linux, and VPS deployment                           |

## ORGANIZATIONAL EXPERIENCE

**Data Science Workshop - ITS x NTUST Taiwan** 2025  
*Committee Member* Surabaya, Indonesia

- Prepared workshop materials and coordinated session delivery for an international data-science collaboration.
- Documented each session and produced Instagram Reels for real-time event coverage, creating an organized archive for post-event reporting.

**Mathematics Students Association (HIMATIKA) ITS - Media and Information** 2025  
*Staff Member* Surabaya, Indonesia

- Planned social media content and communications for the Mathematics Student Association to increase the visibility of its activities
- Organized event documentation and edited videos for promotional and archival use

**Islamic Mathematics Study Institute - Media and Information** 2024 - 2025  
*Division Chief* Surabaya, Indonesia

- Led a 10-person team, coordinated content and documentation workflows, and grew the organization's audience by more than 100 followers through planned communications.
- Developed a comprehensive content strategy and evaluated content performance to improve audience engagement and reach.

**OMITS 17th** 2023 - 2024  
*Coordinator of Website* Surabaya, Indonesia

- Led a four-person development team, directing backend work to maintain and secure the server while guiding frontend implementation to deliver an eye-catching website for more than 2,000 participants.
- Deployed the server and maintained system reliability throughout the event lifecycle.